

Calculus II Lesson 09

1) $a_n = 5^n$

a) $a_1 = \underline{\quad ? \quad}$

b) $a_2 = \underline{\quad ? \quad}$

c) $a_3 = \underline{\quad ? \quad}$

d) $a_4 = \underline{\quad ? \quad}$

e) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 5^n = \underline{\quad ? \quad}$

$$2) a_n = \cos\left(\frac{\pi n}{4}\right)$$

$$a) a_1 = \underline{\quad ? \quad}$$

$$b) a_2 = \underline{\quad ? \quad}$$

$$c) a_3 = \underline{\quad ? \quad}$$

$$d) a_4 = \underline{\quad ? \quad}$$

$$e) \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left[\cos\left(\frac{\pi n}{4}\right) \right] = \underline{\quad ? \quad}$$

$$3) \ a_n = (-1)^n \left(\frac{n}{4} \right)$$

$$a) \ a_1 = \underline{\quad ? \quad}$$

$$b) \ a_2 = \underline{\quad ? \quad}$$

$$c) \ a_3 = \underline{\quad ? \quad}$$

$$d) \ a_4 = \underline{\quad ? \quad}$$

$$e) \ \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left[(-1) \left(\frac{n}{4} \right) \right] = \underline{\hspace{2cm} ? \hspace{2cm}}$$

4) Recursive Sequence.

$$a_1 = 4; \quad a_{n+1} = 2(a_n + 4)$$

a) $a_2 = \underline{\quad ? \quad}$

b) $a_3 = \underline{\quad ? \quad}$

c) $a_4 = \underline{\quad ? \quad}$

d) $a_5 = \underline{\quad ? \quad}$

5) Recursive Sequence.

$$a_1 = 2; \quad a_{n+1} = \frac{2}{3}a_n^3$$

a) $a_2 = \underline{\quad}$

b) $a_3 = \underline{\quad}$

c) $a_4 = \underline{\quad}$

d) $a_5 = \underline{\quad}$

6) Simplify $\frac{(n+2)!}{n!}$

Hint: $n! = (n)(n-1)(n-2)(n-3)(n-4)\cdots(3)(2)(1)$

$$(n+2)! = (n+2)(n+1)(n)(n-1)(n-2)(n-3)(n-4)\cdots(3)(2)(1)$$

a) $\frac{(n+2)!}{n!} = \underline{\hspace{2cm}}?$

b) $\lim_{n \rightarrow \infty} \left[\frac{(n+2)!}{n!} \right] = \underline{\hspace{2cm}}?$

7) Simplify $\frac{(2n+2)!}{(2n-1)!}$

Hint: $(2n+2)! = (2n+2)(2n+1)(2n)(2n-1)(2n-2)(2n-3)\cdots(3)(2)(1)$

$(2n-1)! = (2n-1)(2n-2)(2n-3)(2n-4)(2n-5)\cdots(3)(2)(1)$

a) $\frac{(2n+2)!}{(2n-1)!} = \underline{\hspace{2cm} ? \hspace{2cm}}$

b) $\lim_{n \rightarrow \infty} \left[\frac{(2n+2)!}{(2n-1)!} \right] = \underline{\hspace{2cm} ? \hspace{2cm}}$

8) Simplify $\frac{(2(n+1))!}{(2n)!}$

Hint:

$$(2(n+1))! = (2n+2)! = (2n+2)(2n+1)(2n)(2n-1)(2n-2)\cdots(3)(2)(1)$$

$$(2n)! = (2n)(2n-1)(2n-2)(2n-3)(2n-4)(2n-5)\cdots(3)(2)(1)$$

a) $\frac{(2(n+1))!}{(2n)!} = \underline{\hspace{2cm}} ?$

b) $\lim_{n \rightarrow \infty} \left[\frac{(2(n+1))!}{(2n)!} \right] = \underline{\hspace{2cm}} ?$

9) Let $a_n = \frac{3n^2}{n^2 + 2}$

a) Simplify $\frac{3n^2}{n^2 + 2} = \frac{(3n^2)/n^2}{(n^2 + 2)/n^2} = \frac{?}{?}$ (multiple choice)

A) $\frac{3n}{1 - \frac{2}{n^2}}$

B) $\frac{3n^2}{1 - \frac{2}{n^2}}$

C) $\frac{3}{1 - \frac{2}{n^2}}$

b) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{3n^2}{n^2 + 2} = \frac{?}{?}$

10) Let $a_n = \frac{4n}{\sqrt{4n^2 + 1}}$

a) Simplify $\frac{4n}{\sqrt{4n^2 + 1}} = \frac{(4n)/\sqrt{n^2}}{(\sqrt{4n^2 + 1})/\sqrt{n^2}} = \frac{(4n)/n}{\sqrt{\frac{4n^2}{n^2} + \frac{1}{n^2}}} = ?$ (multiple choice)

A) $\frac{4}{\sqrt{4 + \frac{1}{n^2}}}$

B) $\frac{4n}{\sqrt{4 + \frac{1}{n^2}}}$

C) $\frac{4}{\sqrt{4n + \frac{1}{n^2}}}$

b) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{4n}{\sqrt{4n^2 + 1}} = ?$

11) Let $a_n = \frac{7n+1}{n}$

a) Simplify $\frac{7n+1}{n} = \frac{(7n+1)/n}{(n)/n} = \underline{\hspace{1cm}}$ (multiple choice)

A) $\frac{7n+\frac{1}{n}}{1}$

B) $\frac{7+\frac{1}{n}}{n}$

C) $\frac{7+\frac{1}{n}}{1}$

b) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{7n+1}{n} = \underline{\hspace{1cm}}$

12) Let $a_n = \cos \frac{n\pi}{2}$

a) $\cos\left(\frac{1\pi}{2}\right) = ?$

b) $\cos\left(\frac{2\pi}{2}\right) = ?$

c) $\cos\left(\frac{3\pi}{2}\right) = ?$

d) $\cos\left(\frac{4\pi}{2}\right) = ?$

e) $\cos\left(\frac{5\pi}{2}\right) = ?$

f) $\cos\left(\frac{6\pi}{2}\right) = ?$

g) $\cos\left(\frac{7\pi}{2}\right) = ?$

h) $\cos\left(\frac{8\pi}{2}\right) = ?$

i) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \cos \frac{n\pi}{2} = ?$

13) Let $a_n = \frac{\sqrt[3]{n+1}}{\sqrt[3]{n}}$

a) Simplify $\frac{\sqrt[3]{n+1}}{\sqrt[3]{n}} = \frac{\sqrt[3]{n+1} / \sqrt[3]{n}}{\sqrt[3]{n} / \sqrt[3]{n}} = \frac{\sqrt[3]{n/n+1/n}}{\sqrt[3]{n} / \sqrt[3]{n}} = \underline{\hspace{1cm}}$ (multiple choice)

A) $\frac{\sqrt[3]{1+1/n}}{1}$

B) $\frac{\sqrt[3]{n+1/n}}{1}$

C) $\frac{\sqrt[3]{1+1/n^3}}{1}$

b) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n+1}}{\sqrt[3]{n}} = \underline{\hspace{1cm}}$

14) Let $a_n = \frac{3n^2 + 5n + 7}{5n^2 - 6}$

a) Simplify $\frac{3n^2 + 5n + 7}{5n^2 - 6} = \frac{(3n^2 + 5n + 7)/n^2}{(5n^2 - 6)/n^2} = \underline{\hspace{2cm}}$ (multiple choice)

A) $\frac{3n + 5/n + 7/n^2}{5n - 6/n^2}$

B) $\frac{3n^2 + 5/n + 7/n^2}{5n^2 - 6/n^2}$

C) $\frac{3 + 5/n + 7/n^2}{5 - 6/n^2}$

b) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{3n^2 + 5n + 7}{5n^2 - 6} = \underline{\hspace{2cm}}$

